Team SCADA

Project Sponsor: POWER Engineers, Inc

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Project Background

Role of SCADA in Power Distribution:

- Supervisory Control and Data Acquisition (SCADA) is a system that acts as the central hub for collecting, processing, and visualizing data from field devices

- SCADA provides real-time monitoring, fault detection, data acquisition and remote control of field devices such as:
  - Relays, meters, breaker controllers, and other Intelligent Electronic Devices (IEDs)

- SCADA enables utilities to:
  - Perform automated switching and load management
  - Log and visualize data for compliance and analysis
  - Enhance security and grid stability

Project Background

Intelligent Electronic Devices (IEDs):

- IEDs combine measurement, communication, and protection, in one package
    - Ex: SEL-315S, SEL-735, SEL-587Z, SEL-487V, GE-L90
- IEDs present key challenges:
    - High cost: Individual units can cost around $2,000-$12,000
    - Bulky Form Factor: Require panels, wiring, and auxiliary power
    - Limited Scalability: Constrained by physical space and hardware limitations

Project Motivation

Challenges in SCADA Simulation

High Costs:

- Hardware, software, and licensing fees can be cost prohibitive

Limited Scalability:

- Traditional test sets are difficult to reconfigure or expand without major investments

Complex Setup:

- Requires dedicates racks, wiring, power, training

Our Solution Low cost IED simulator

Low Cost:

- Roughly $140 to simulate 10 devices

Scalable:

- Can handle up to 500 registers

Modular System:

- Minimal footprint solution that is portable easily configurable

Objectives

- Design and develop a low-cost device to emulate substation IEDs to be polled by a designated remote terminal unit (RTU) which would be an SEL real time automation controller (RTAC) using MODBUS
- Simulate multiple devices at once with scalable emulators.
- Develop and produce modular housing for each device, ensuring adaptability and ease of configuration within the lab environment.
- Provide a user interface with the ability to configure each emulator to a specific device.

Project Scope and Budget

Deliverables:

- IED simulator device
- User manual guide
- Sample Config files
- Comprehensive documentation

Budget:

- We have been given a budget of $1000 from our sponsor

Budget

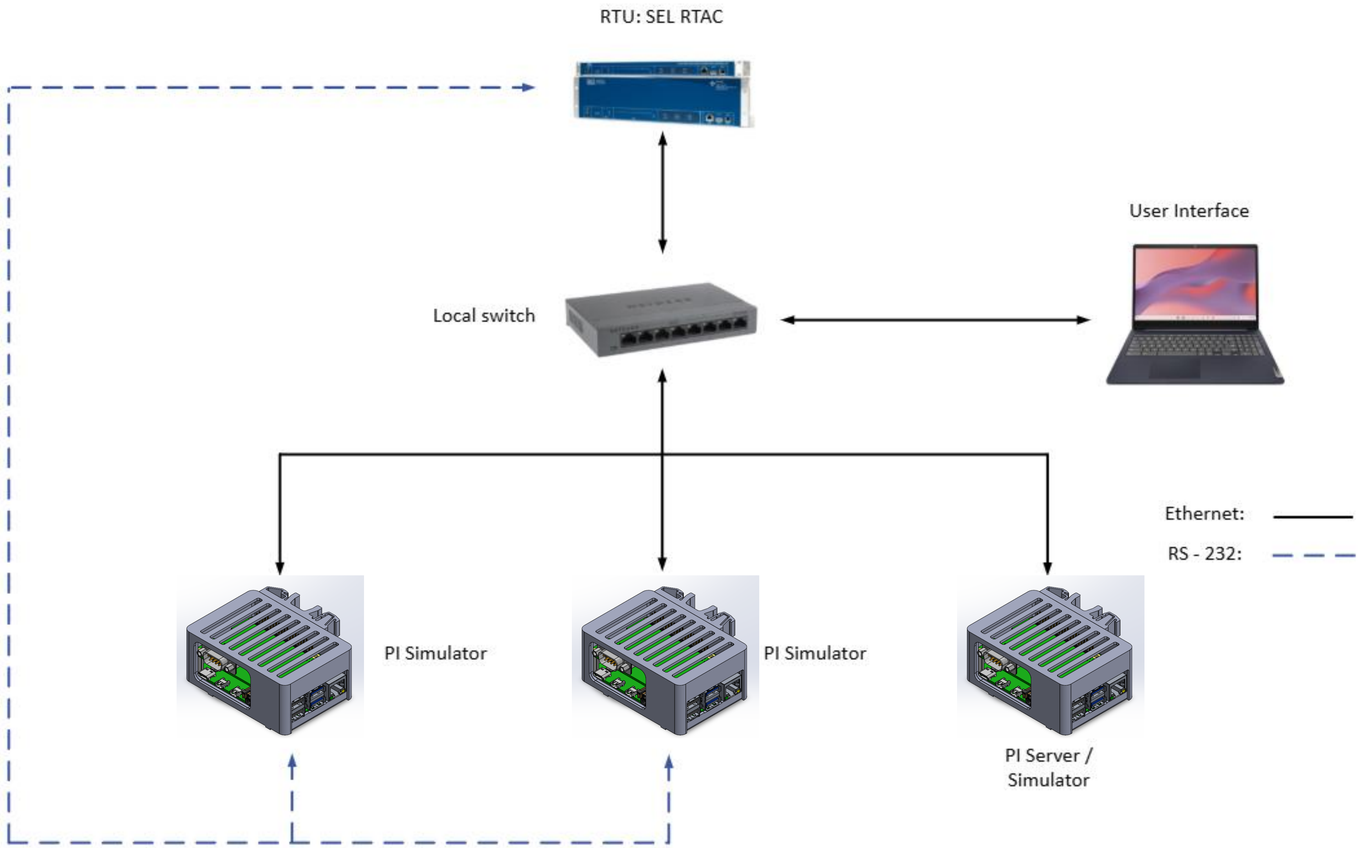
Current Purchases

Item	Cost	Quantity	Total
Raspberry Pi 4 Model B 2GB	$56.99	1	$56.99
Power Supply	$8.77	1	$8.77
PI-232 RS232 Hat	$18.90	1	$18.90
Network Switch	$17.99	1	$17.99
256GB Micro SD Card	$19.99	1	$19.99
PETG Filament	$15.99	1	$15.99
		Total Spent	$138.63
		Budget Remaining	$861.37

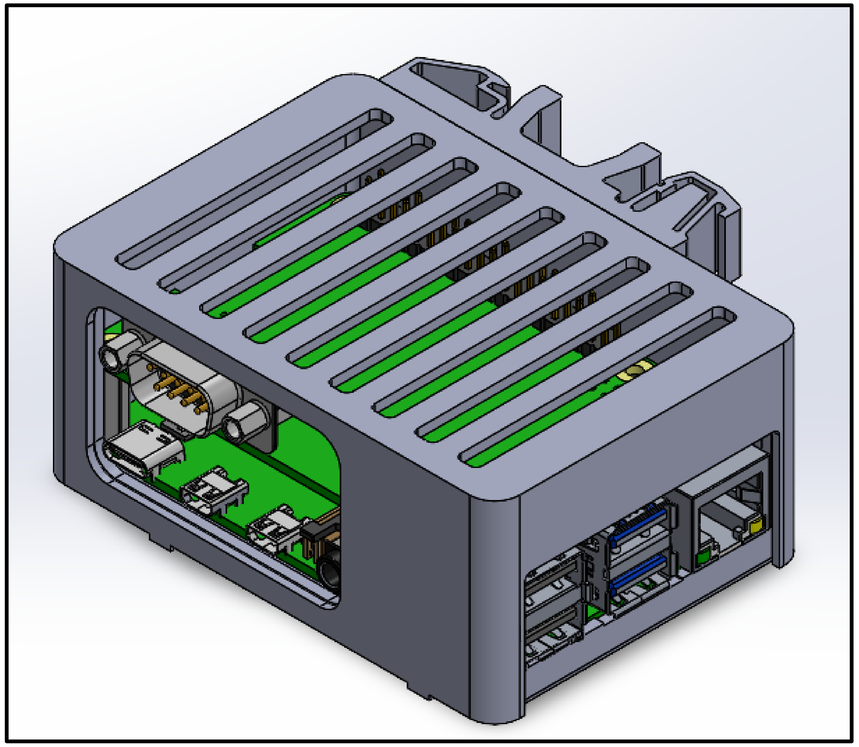
Parts to Scale System

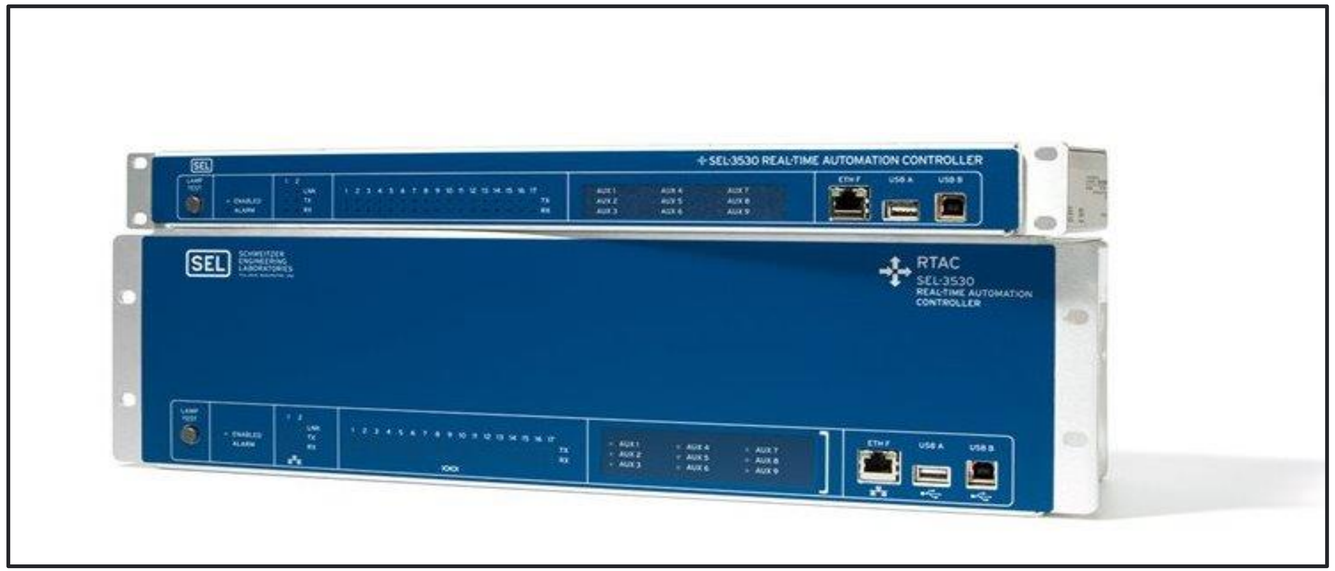
Item	Cost	Quantity	Total
Raspberry Pi 4 Model B 2GB	$56.99	1	$56.99
Power Supply	$8.77	1	$8.77
PI-232 RS232 Hat	$18.90	1	$18.90
256GB Micro SD Card	$19.99	1	$19.99
Miscellaneous	$10.00		$10.00
		Price Per Additional Pi	$114.65

Proposed Solutions



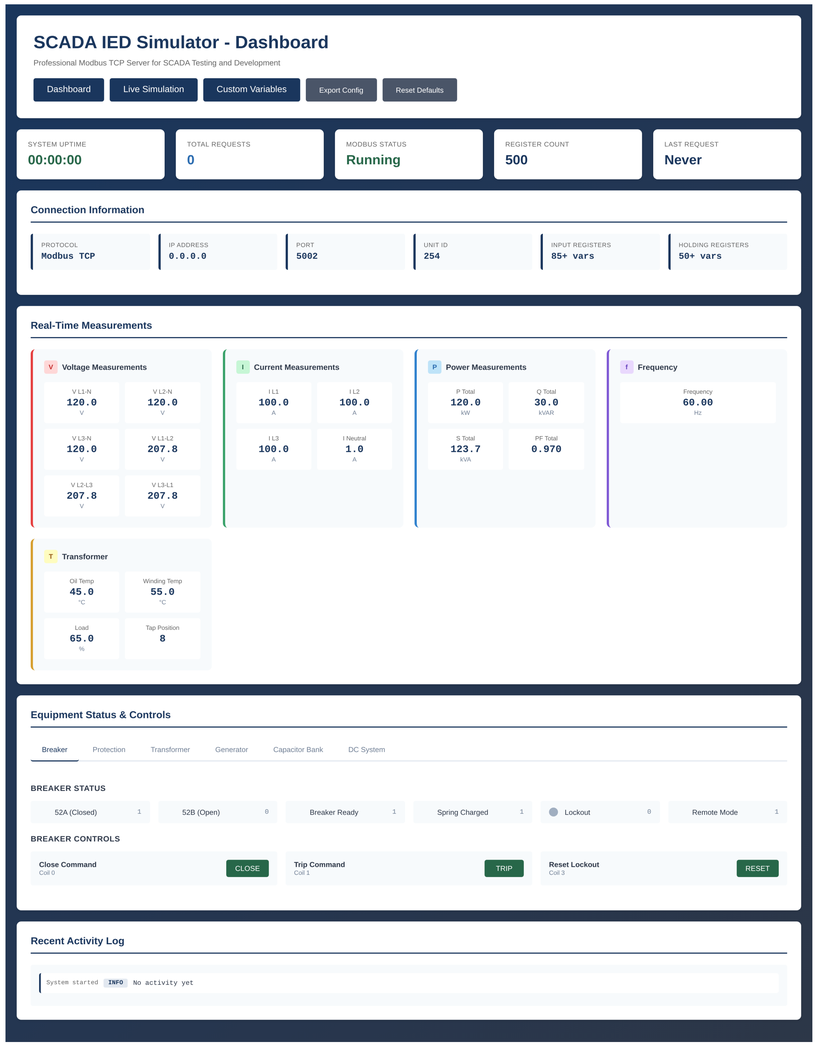
Preliminary Testing and Prototypes





Flask Web Server Output

- Flask runs a local web server on port 5000
- Configuration is retrieved from scada_config.json file, served as JSON
- The browser image confirms proper routing and JSON rendering
- Accessible from the local network at; http://*serverIP*:5000



SCADA Modbus Sim (Enhanced_app.py)

- Heart of the simulator
- Built using pymodbus
- Initializes 500 registers per variable type
  - (Coils, Discrete Inputs, Holding Registers, Input Registers)
- Handles all the variable declarations
  - 260+ pre-configured variables

Category	Variables	Description
Electrical Measurements	50+	Three-phase voltage, current, power, energy
Protection Elements	40+	Overcurrent, voltage, frequency protection status
Transformer Monitoring	15+	Temperature, DGA, tap changer, cooling
Generator Monitoring	15+	MW/MVAR output, excitation, temperatures
Breaker/Switch Status	15+	52A/B, disconnect switches, lockout
Capacitor Bank	10+	Step status, VAR output, controls
DC System	10+	Battery voltage, charger status
Environmental	6	Temperature, humidity, wind

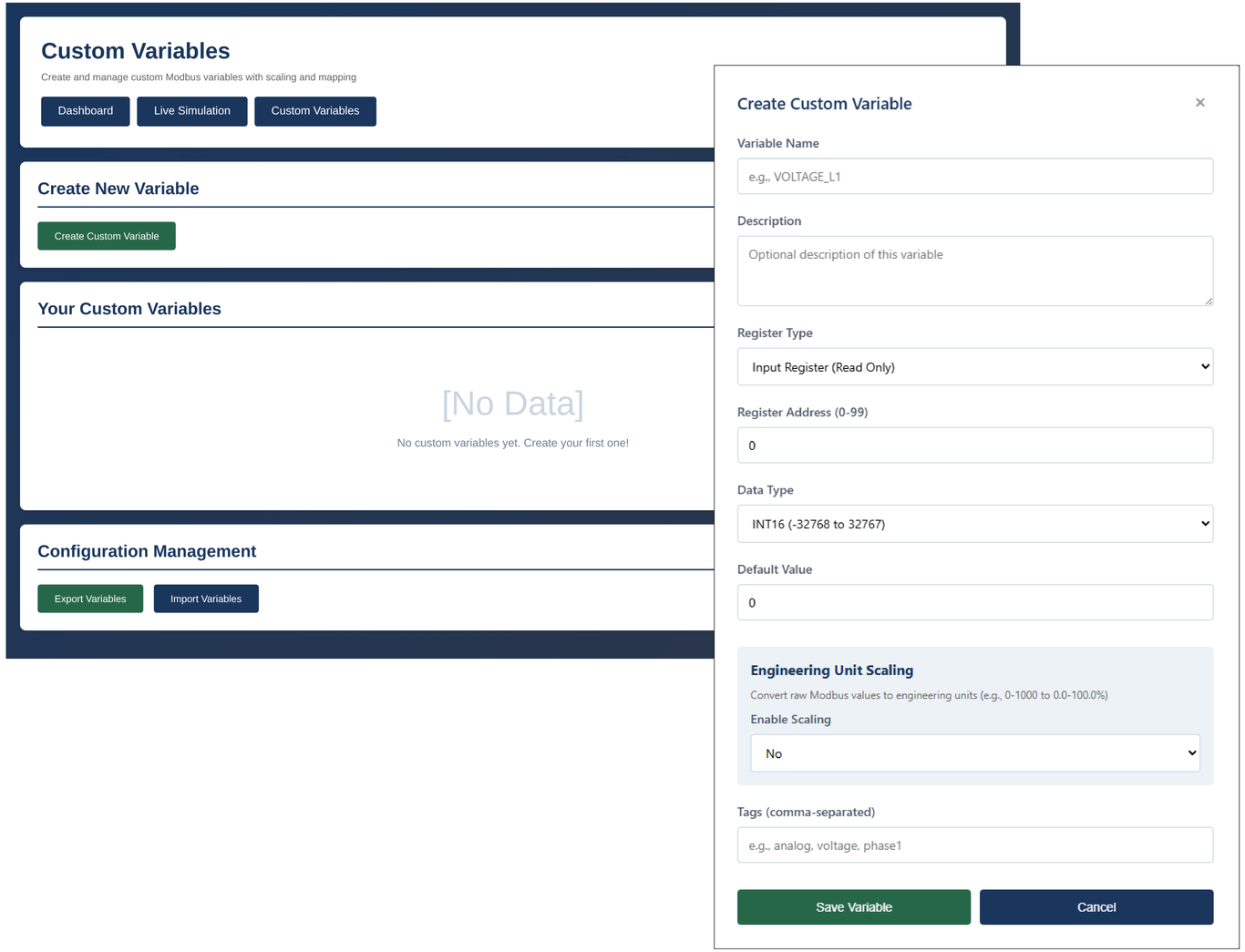
Inside the Raspberry Pi

- Raspberry Pi 4 running
  - Raspberry Pi OS 64-bit
- SSH used to display Flask app
- Flask app successfully running and serving
- Server confirms accessible endpoint and port
- Runs on local network host0.0.0.0, port 5000

```
SCADA@SCADApi: ~/scada_p

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\twsch> ssh SCADA@SCADApi
SCADA@scadapi's password:
Linux SCADApi 6.12.20+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.20-1+rpt1~bpo12+1 (2025-03-19) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Apr  8 18:33:53 2025
SCADA@SCADApi:~ $ ls
Bookshelf  Desktop  Documents  Downloads  modbus_env  Music  Pictures  Public  scada_project  Templates  Videos
SCADA@SCADApi:~ $ cd scada_project
SCADA@SCADApi:~/scada_project $ ls
app.py  config_reader.py  modbus_sim.py  scada_config  venv
SCADA@SCADApi:~/scada_project $ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.254.66:5000
Press CTRL+C to quit
192.168.254.17 - - [24/Apr/2025 12:18:44] "GET / HTTP/1.1" 200 -
192.168.254.17 - - [24/Apr/2025 12:18:45] "GET /favicon.ico HTTP/1.1" 404 -
```

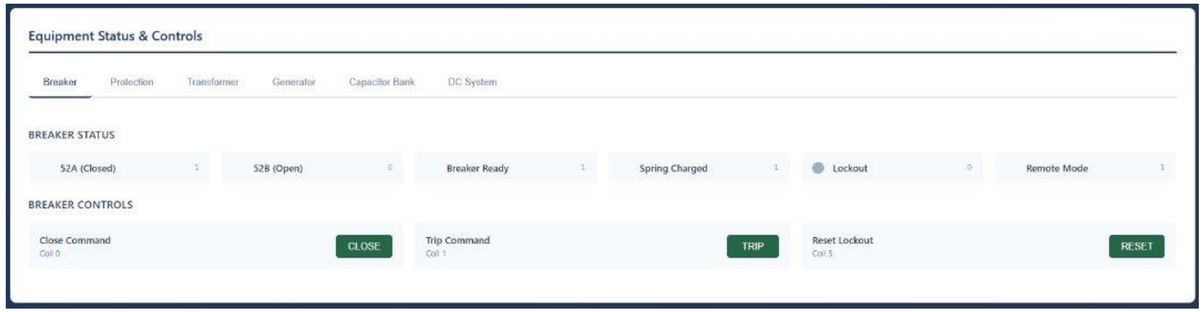
SCADA Variables

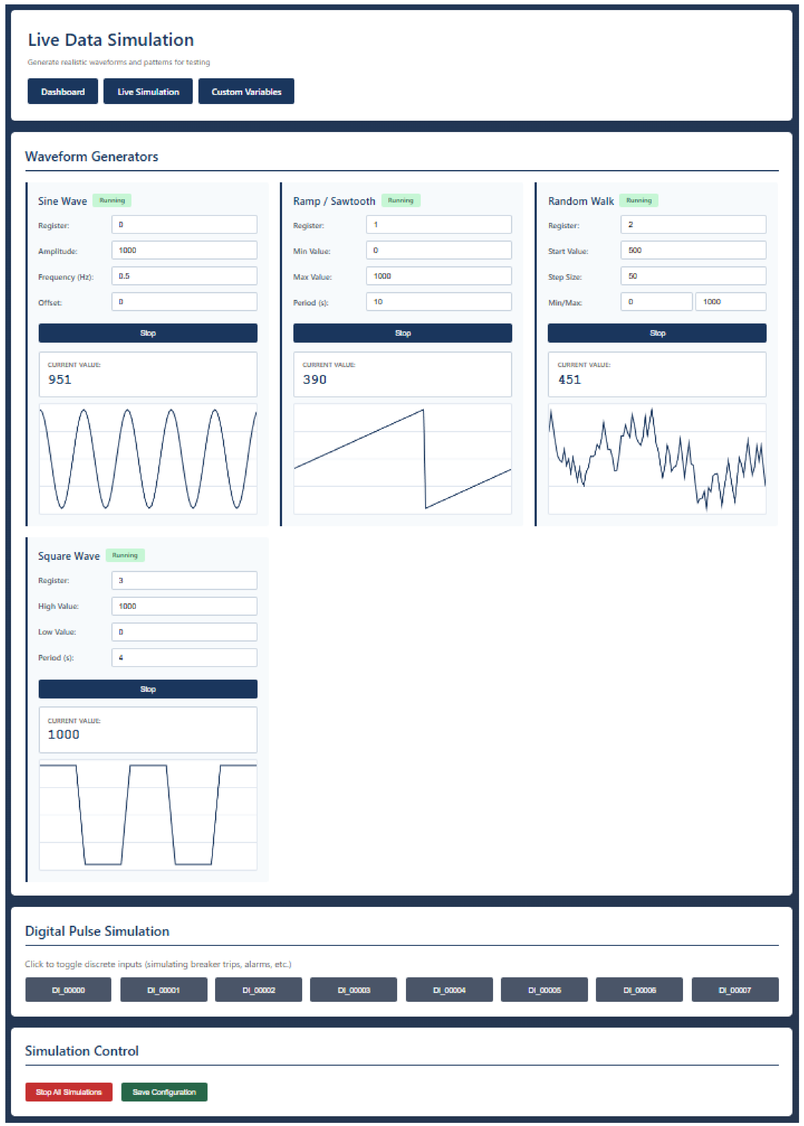
- Create custom variables
- Import/Export Variables
- Specify Register type and address
- Allows for Engineer Unit Scaling



Modbus Simulation

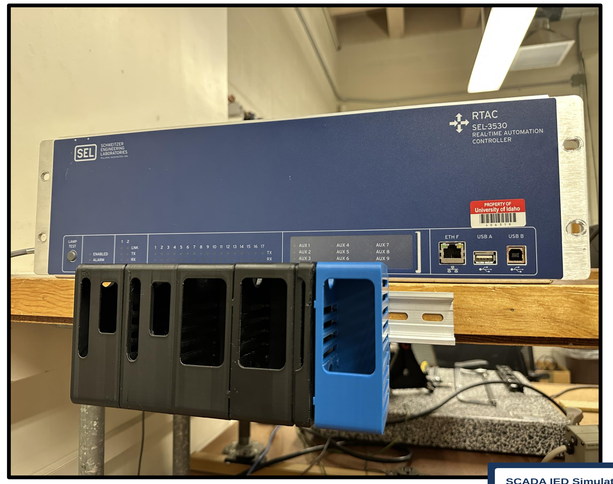
- 260+ Power Industry Variables
- Advanced Search & Filtering
- Organized by Category
- Export Functionality
- Real-time Statistics

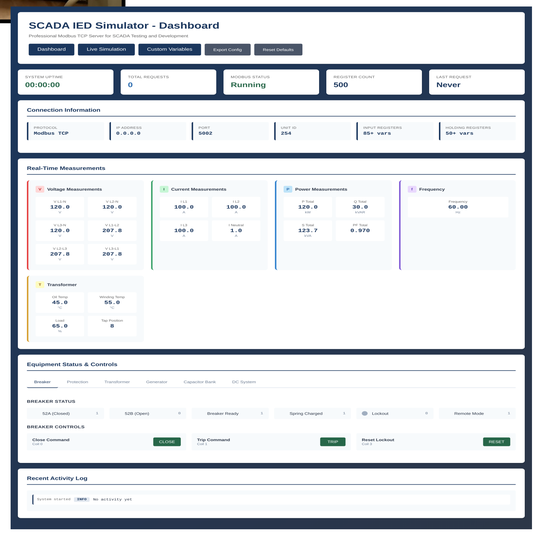




Milestones

- Developed IED PI simulator that can simulate analog and digital data, as well as counters.
- Finalized Pi housing design that is compatible with DIN rail
- Used SEL RTAC for polling data from the PI simulator
- Finalized HMI
- Developed user manual guide that a user can follow to configure PI simulator based on the user needs





Results

- Full Modbus TCP server (function codes 01–06, 15, 16)
- 500 registers per type (Coils, Discrete Inputs, Holding, Input Registers)
- 260+ pre-configured power industry variables
- Web-based HMI with real-time dashboard and live simulation
- REST API for programmatic control
- Configuration import/export
- Runs on Raspberry Pi 4 or any Linux system
- Github Repository with easy install and user guide

Github Repository

- https://github.com/TWi5td/CAPSTONE_SCADA_sim_pi
- Source code
- User Manual
- Easy install
- Troubleshooting

Questions???